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Higher-order moments of generalized polynomial chaos expansions for intrusive and non intrusive uncertainty quantification

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In this paper we summarize the main results available for the computation of higher-order moments of continuous hypergeometric polynomials of the Askey family of orthogonal polynomials. Specifically, we provide with analytical formulas for the third-order moments of Jacobi, generalized Hermite, and generalized Laguerre polynomials, together with simple `Matlab` codes implementing them. Fourth-order moments and beyond are obtained straightforwardly by induction. As for applications of these results we have in mind the polynomial chaos method for uncertainty quantification in computational structural mechanics and computational fluid mechanics, among other possible fields of research. Generalized polynomial chaos expansions are used either intrusively or non intrusively in the available numerical methods and codes to propagate parametric uncertainties. Higher-order moments of the underlying polynomials are strictly needed in the former approach, while they are useful in both approaches as long as higher-order moments of the quantities of interest need be post-processed in the analysis.

I. Introduction

The polynomial chaos (PC), or homogeneous chaos expansion consists in the decomposition of a second-order random variable in a series of multivariate Hermite polynomials in a countable sequence of independent Gaussian random variables [6,36]. Specifically, it considers truncations of such an expansion for the purpose of constructing approximations of the foregoing second-order random variable. It has been intensively used in recent years in computational methods for solving ordinary or partial differential equations with random data or parameterized by Gaussian random variables. As the solutions of these equations are stochastic processes indexed by spatial and/or time coordinates, which are typically second-order random fields from physical considerations (they have finite energy), homogeneous chaos expansions are used to approximate them along the stochastic dimension. For random fields parameterized by non Gaussian, arbitrary random variables the numerical study in [37] has shown that the convergence rates of Hermite polynomial chaos are not optimal. This observation has prompted the development of generalized polynomial chaos expansions (gPC) involving other families of polynomials which are orthogonal with respect to the probability density function (PDF) of the random parameters [31, 37]. Optimal, namely exponential rates of convergence are achieved once this substitution has been done.

The earlier approach of using homogeneous chaos to numerically solve differential equations proposed truncated PC expansions as trial functions in a Galerkin formulation, resulting in the spectral stochastic finite element method, or stochastic Galerkin method [33] subsequently developed by Ghanem & Spanos [16]. More precisely, the chaos expansions of the model parameters and variables are substituted in the model equations, which in turn yield evolution equations for the expansion coefficients of the solution processes

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from Galerkin projections using the orthogonal polynomials of the PC expansion as test functions. This procedure gives rise to third-order, indeed fourth-order, or even sixth-order moments of these polynomials as illustrated by some examples in [11, 12, 15, 16, 21–23, 25]. They are typically evaluated by Gauss quadratures in most applications, and then stored for use throughout the computations. The stochastic Galerkin method is intrusive in that it may require significant alterations of the existing deterministic codes used for simulating structural or fluid dynamics. Generally it also yields coupled equations for the expansion coefficients.

This situation has prompted the development of non intrusive approaches, which require only repetitive executions of existing deterministic codes. Stochastic collocation methods [4, 17, 22, 34, 38, 39] have become widely used in computational fluid dynamics (CFD), for the applicability of these uncertainty quantification techniques is not affected by the complexity and high nonlinearity of the existing flow solvers so long as they achieve a reasonable accuracy. Complex aerodynamic analysis and design of aircraft use such high-fidelity CFD tools for shape optimization for example, whereby some robustness is achieved by considering uncertain operational, environmental, or manufacturing parameters. The objective functions in these optimization processes are often expressed in terms of moments of the output quantities of interest, such as their means, standard deviations, or even higher-order statistics (skewness, kurtosis...). PC and gPC expansions can be used to approximate non intrusively these quantities of interest, and then compute their moments. It should be noted at this stage that such post-processing steps also pertains to the intrusive stochastic Galerkin method so long as higher-order statistics of the random solutions are requested.

The main purpose of this paper is to summarize the main analytical formulas available for these moments, so that they could be advantageously evaluated "on-the-fly" in the course of the computations. A numerical implementation is also proposed in the form of freely available `Matlab` codes. It is believed that such results may have some relevance for either the intrusive or the non intrusive approaches of uncertainty quantification based on PC or gPC expansions. The paper is organized as follows. The standard linearization problem of a product of polynomials is briefly presented in the next section. Then it is applied to the computation of higher-order moments of orthogonal polynomials in Sect. III, where the available explicit expressions of the third-order moments are listed for Jacobi, generalized Hermite, and generalized Laguerre polynomials (thus covering all continuous polynomials identified in [3, 10, 37] for example). Numerical implementation of these results using `Matlab` is addressed in Sect. IV. It is validated by comparisons with classical evaluations of the third-order moments by Gauss quadratures, for which the codes used in this process are also provided. Some conclusions and perspectives are finally drawn in Sect. V.

II. Standard linearization problem

Let $Q_j(x)$ and $R_k(x)$ be two polynomials of degrees j and k respectively. Let $\{P_n\}_{n\geq 0}$ be an arbitrary sequence of polynomials such that $\deg P_n = n$. The general linearization problem consists in finding the coefficients $B_n(j,k)$ such that:

$$Q_j(x)R_k(x) = \sum_{n=0}^{j+k} B_n(j,k)P_n(x)\,. \tag{1}$$

A particular case of this problem is the standard linearization problem (or Clebsch-Gordan-type problem) for which $Q_n \equiv R_n \equiv P_n$:

$$P_j(x)P_k(x) = \sum_{n=0}^{j+k} B_n(j,k)P_n(x)\,. \tag{2}$$

Another particular case is the so-called connection problem, for which $R_k(x) = 1$; if in addition $Q_j(x) = x^j$ is chosen, it is referred to as the inversion problem for the sequence $\{P_n\}_{n\geq 0}$. These problems have been the subject of numerous investigations, some of them being addressed in [1, 2, 5, 7–9, 13, 14, 19, 26, 28, 35] and references therein. The objective of this communication is definitely not to review exhaustively these results, but to apply them to the computation of higher-order moments from PC or gPC expansions of random parameters and/or functionals.

III. Higher-order moments of orthonormal polynomials

Consider now the standard linearization problem for the family $\{P_n\}_{n\geq 0}$ of orthogonal polynomials with respect to the non-negative density $x \mapsto \mu(x)$ of support I, *i.e.*:

$$\langle P_m P_n\rangle_\mu := \int_I P_m(x)P_n(x)\mu(x)\mathrm{d}x = \gamma_n\delta_{mn}\,, \quad m,n\geq 0\,, \tag{3}$$

where δ_{mn} is the usual Kronecker symbol, and $\gamma_n > 0$ is the normalization constant. Then clearly from Eq. (2) the following holds:

$$\langle \hat{P}_j\hat{P}_k\hat{P}_l\rangle_\mu = \sqrt{\frac{\gamma_l}{\gamma_j\gamma_k}}B_l(j,k) := b_l(j,k)\,, \quad l\leq j+k\,, \tag{4}$$

if we introduce the orthonormalized polynomials $\hat{P}_n \equiv \gamma_n^{-\frac{1}{2}}P_n$. The roles of j,k and l in Eq. (4) are transparent so they can be permutated in this formula. This should be apparent in the analytical expression of $b_l(j,k)$ whenever it is available. In addition, one has $b_l(j,k)=0$ whenever $l<|j-k|$. Indeed, either $k+l<j$ or $j+l<k$ in this case thus $\deg\{P_kP_l\}<\deg\{P_j\}$ or $\deg\{P_jP_l\}<\deg\{P_k\}$, and consequently $\langle P_jP_kP_l\rangle_\mu = 0$. The fourth-order moment can be derived from the above third-order moments by simple mathematical induction:

$$\begin{aligned}\langle \hat{P}_j\hat{P}_k\hat{P}_l\hat{P}_m\rangle_\mu &= \sum_{n=0}^{j+k} b_n(j,k)\langle \hat{P}_l\hat{P}_m\hat{P}_n\rangle_\mu \\ &= \sum_{n=0}^{j+k} b_n(j,k)b_n(l,m)\,.\end{aligned} \tag{5}$$

Likewise, free permutations of the transparent indices j,k,l and m are applicable. Higher-order moments are obtained along the same lines by repeated uses of Eq. (2) and induction. The above third-order and fourth-order moments of orthonormal polynomials typically arise in the determination of the PC expansion for the product of two or three stochastic variables, as illustrated in [12] for example. Here the third order tensor $C_{jkl} = \langle P_jP_kP_l\rangle_\mu$ is rather evaluated numerically by dedicated quadrature rules, benefiting to some extent from its sparsity. The foregoing analysis rather shows that they can be obtained from the linearization coefficients of the corresponding standard linearization problem. These coefficients are now explicitly given for some classical families of orthogonal polynomials of the Askey scheme [3] in the subsequent sections. Jacobi, generalized Hermite, and generalized Laguerre polynomials are more particularly addressed. Families corresponding to discrete non-negative measures $\mu(\mathrm{d}x)$ may be considered alike, though they are not reviewed in this communication.

A. Jacobi polynomials

The Jacobi polynomials $\{J_n^{(\alpha,\beta)}\}_{n\geq 0}$ are orthogonal with respect to the weight function $\mu(x) = (1-x)^\alpha(1+x)^\beta$, with $\alpha,\beta > -1$ and $I=[-1,1]$. They are defined by *e.g.* the standard Rodrigues' formula:

$$J_n^{(\alpha,\beta)}(x) = \frac{1}{\mu(x)}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(\frac{\mu(x)}{n!}\left(\frac{x^2-1}{2}\right)^n\right) = \sum_{j=0}^{n}\binom{n+\alpha}{n-j}\binom{n+\beta}{j}\left(\frac{x-1}{2}\right)^j\left(\frac{x+1}{2}\right)^{n-j}\,,$$

where:

$$\binom{z}{p} = \frac{z!}{(z-p)!p!}$$

stands for the generalized binomial coefficient. Indeed, one has $z! := \Gamma(z+1) = \int_0^{+\infty} t^z\mathrm{e}^{-t}\mathrm{d}t$ and $\Gamma(p+1)=p!$, the usual factorial, if p is an integer. Jacobi polynomials arise in gPC expansions for random variables following beta distributions of the first kind; see *e.g.* [29,37]. The normalization constant γ_n in Eq. (3) then reads:

$$\gamma_n = \frac{2^{\alpha+\beta+1}(n+\alpha)!(n+\beta)!}{(2n+\alpha+\beta+1)(n+\alpha+\beta)!n!}\,.$$

The linearization coefficients $B_n(j,k)$ in the general linearization problem:

$$J_j^{(\lambda,\delta)}(x)J_k^{(\mu,\nu)}(x) = \sum_{n=|j-k|}^{j+k} B_n(j,k)J_n^{(\alpha,\beta)}(x)\,, \quad \alpha,\beta,\lambda,\delta,\mu,\nu > -1\,,$$

are given in [8, Eq. (12)] in terms of double hypergeometric functions (the so-called Kampé de Fériet functions). In the context of PC expansions we are rather interested in the standard linearization problem for which $\alpha=\lambda=\mu$ and $\beta=\delta=\nu$. A representation in terms of a terminating generalized hypergeometric series was also derived in [1,26] for this problem. In [8] the linearization coefficients for the standard problem read:

$$\begin{aligned} B_{j+k-n}(j,k) = \frac{(-1)^n(j+k)!}{j!k!n!} & \frac{(\alpha+1)_{j+k}(\alpha+\beta+1)_{2j}(\alpha+\beta+1)_{2k}}{(\alpha+\beta+1)_j(\alpha+\beta+1)_k} \\ & \times \frac{(2(j+k-n)+\alpha+\beta+1)(\alpha+\beta+1)_{j+k-n}}{(\alpha+1)_{j+k-n}(\alpha+\beta+1)_{2(j+k)-n+1}} \\ & \times F_{2:1}^{2:2}\left(\left.\begin{matrix}[-n,-\alpha-\beta-1-2(j+k)+n]:[-j,-\alpha-j];[-k,-\alpha-k] \\ [-(j+k),-\alpha-(j+k)]:[-2j-\alpha-\beta];[-2k-\alpha-\beta]\end{matrix}\right| 1,1\right)\,, \end{aligned} \tag{6}$$

where $(z)_n := \frac{\Gamma(z+n)}{\Gamma(z)}$ stands for the usual Pochhammer symbol, and $F_{m:n}^{p:q}$ is the Kampé de Fériet function defined by:

$$F_{m:n}^{p:q}\left(\left.\begin{matrix}(\boldsymbol{a}_p):(\boldsymbol{b}_q);(\boldsymbol{c}_q) \\ (\boldsymbol{d}_m):(\boldsymbol{e}_n);(\boldsymbol{f}_n)\end{matrix}\right| x,y\right) = \sum_{k=0}^{\infty}\sum_{l=0}^{\infty}\frac{[\boldsymbol{a}_p]_{k+l}[\boldsymbol{b}_q]_k[\boldsymbol{c}_q]_l}{[\boldsymbol{d}_m]_{k+l}[\boldsymbol{e}_n]_k[\boldsymbol{f}_n]_l}\frac{x^k}{k!}\frac{y^l}{l!}\,,$$

with $[\boldsymbol{a}_p]_k = \prod_{j=1}^p (a_j)_k$. For numerical robustness, we will rather resort to the older induction formula derived in [14]. Here the linearization coefficients are given by:

$$B_n(j,k) = (-1)^{j+k+n}\frac{k!}{(k+\beta)!}\tilde{B}_n(j,k)\,, \tag{7}$$

where the coefficients $\tilde{B}_n(j,k)$ are obtained by the induction formula [19, Eq. (4.13)]:

$$\begin{aligned} & \frac{[(j+k+\alpha+\beta+1)^2-(n+\alpha+\beta)^2][(n+\alpha+\beta)^2-(j-k)^2]}{(2n+\alpha+\beta)[2(n-1)+\alpha+\beta+1]}(n+\beta)\tilde{B}_{n-1}(j,k) \\ & \quad - \frac{[(j+k+\alpha+\beta+1)^2-(n+1)^2][(n+1)^2-(j-k)^2]}{[2(n+1)+\alpha+\beta][2(n+1)+\alpha+\beta+1]}(n+1+\alpha)\tilde{B}_{n+1}(j,k) \\ & \quad + \frac{[(j+k+\alpha+\beta+1)^2-(n+1+\alpha+\beta)^2][(n+1)^2-(j-k)^2]}{2(n+1)+\alpha+\beta}\left(\frac{\beta-\alpha}{2}\right)\tilde{B}_n(j,k) \\ & \quad - \frac{[(j+k+\alpha+\beta+1)^2-(n+\alpha+\beta)^2][n^2-(j-k)^2]}{2n+\alpha+\beta}\left(\frac{\beta-\alpha}{2}\right)\tilde{B}_n(j,k) = 0\,, \end{aligned}$$

starting from (assuming $j \geq k$):

$$\tilde{B}_{j-k-1}(j,k) = 0\,, \quad \tilde{B}_{j-k}(j,k) = \frac{[2(j-k)+\alpha+\beta+1]!(2k+\alpha+\beta)!(j+\alpha)!(j+\beta)!}{(2j+\alpha+\beta+1)!(k+\alpha+\beta)!(j-k+\alpha)!(j-k)!j!}\,.$$

The ultraspherical (Gegenbauer) polynomials $\{C_n^{(\lambda)}\}_{n\geq 0}$ correspond to the particular case $\alpha=\beta=\lambda-\frac{1}{2}$ with $\lambda \neq 0$ and the standardization:

$$C_n^{(\lambda)}(x) = \frac{(2\lambda)_n}{(\lambda+\frac{1}{2})_n}J_n^{(\lambda-\frac{1}{2},\lambda-\frac{1}{2})}(x)\,, \quad \lambda > -\frac{1}{2}\,.$$

The corresponding linearization coefficients $B_n(j,k)$ such that:

$$C_j^{(\lambda)}(x)C_k^{(\lambda)}(x) = \sum_{n=0}^{\min(j,k)} B_{j+k-2n}(j,k)C_{j+k-2n}^{(\lambda)}(x)\,, \quad \lambda > -\frac{1}{2}\,,$$

are given by the Dougall's formula [3, Eq. (5.7)] (see also [8, Eq. (28)]):

$$B_{j+k-2n}(j,k)=\frac{(\lambda+j+k-2n)(j+k-2n)!}{(\lambda+j+k-n)n!(j-n)!(k-n)!}\frac{(2\lambda)_{j+k-n}(\lambda)_{j-n}(\lambda)_{k-n}(\lambda)_n}{(2\lambda)_{j+k-2n}(\lambda)_{j+k-n}}\,. \tag{8}$$

For the family of Legendre polynomials such that $\lambda=\frac{1}{2}$ the Neumann-Adams formula [2, 24, p. 91] is recovered, namely:

$$B_{j+k-2n}(j,k)=\frac{2(j+k-2n)+1}{2(j+k-n)+1}\frac{(j+k-n)!(\frac{1}{2})_{j-n}(\frac{1}{2})_{k-n}(\frac{1}{2})_n}{(\frac{1}{2})_{j+k-n}(j-n)!(k-n)!n!}\,,$$

where $(\frac{1}{2})_n=\frac{(2n)!}{4^n n!}$, *etc.* Legendre polynomials arise in gPC expansions for the important case of uniform distributions.

Lastly, Chebyshev polynomials of the first kind $\{T_n\}_{n\geq 0}$ correspond to the special case $\alpha=\beta=-\frac{1}{2}$ and are:

$$T_n(x)=\cos(n\arccos x)=\frac{J_n^{(-\frac{1}{2},-\frac{1}{2})}(x)}{J_n^{(-\frac{1}{2},-\frac{1}{2})}(1)}\,,\quad J_n^{(-\frac{1}{2},-\frac{1}{2})}(1)=\frac{1}{n!}\left(\frac{1}{2}\right)_n\,.$$

Since:

$$2T_j(x)T_k(x)=T_{|j-k|}(x)+T_{j+k}(x)\,,$$

the linearization coefficients are simply $B_{|j-k|}(j,k)=B_{j+k}(j,k)=\frac{1}{2}$ and $B_n(j,k)=0$ otherwise.

B. Hermite polynomials

The generalized Hermite polynomials $\{H_n^{(\alpha)}\}_{n\geq 0}$ are orthogonal with respect to the weight function $\mu(|x|)$ with $\mu(x)=x^{2\alpha}\mathrm{e}^{-x^2}$, $\alpha>-\frac{1}{2}$, on $I=\mathbb{R}$. They are given by the Rodrigues-like formula [10, p. 157]:

$$H_n^{(\alpha)}(x)=\frac{1}{\mu(x)}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(\mu(x)x^nK_n^{(\alpha)}(x)\right)\,, \tag{9}$$

where:

$$K_{2m}^{(\alpha)}(x)=\frac{(-1)^m}{(\alpha+1)_m}\,{}_1F_1\left(\left.\begin{matrix}m\\ m+\alpha+1\end{matrix}\right|x^2\right)\,,$$

$$K_{2m+1}^{(\alpha)}(x)=\frac{(-1)^m}{(\alpha+1)_{m+1}}x\,{}_1F_1\left(\left.\begin{matrix}m+1\\ m+\alpha+2\end{matrix}\right|x^2\right)\,,$$

and ${}_pF_q$ is the generalized hypergeometric function defined as:

$${}_pF_q\left(\left.\begin{matrix}(\boldsymbol{a}_p)\\ (\boldsymbol{b}_q)\end{matrix}\right|x\right)=\sum_{k=0}^{\infty}\frac{(a_1)_k(a_2)_k\cdots(a_p)_k}{(b_1)_k(b_2)_k\cdots(b_q)_k}\frac{x^k}{k!}\,.$$

The normalization constant γ_n in Eq. (3) reads [10, p. 157]:

$$\gamma_n=2^{2n}\left\lfloor\frac{n}{2}\right\rfloor!\left(\left\lfloor\frac{n+1}{2}\right\rfloor+\alpha-\frac{1}{2}\right)!\,,$$

where $\lfloor\cdot\rfloor$ is the largest integer function. This family reduces to the classical Hermite polynomials $\{H_n\}_{n\geq 0}$ for $\alpha=0$. The latter arise in PC expansions for random variables following Gaussian distributions and are the original polynomial chaoses of the stochastic finite element method introduced in [16, 20, 33]. Rodrigues' formula (9) for $\alpha=0$ and $\mu(x)=\mathrm{e}^{-x^2}$ reads:

$$H_n(x)=\frac{1}{\mu(x)}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left((-1)^n\mu(x)\right)=n!\sum_{j=0}^{\lfloor\frac{n}{2}\rfloor}\frac{(-1)^j(2x)^{n-2j}}{(n-2j)!j!}\,,$$

and the normalization constant is (owing to $(-\frac{1}{2})!=\sqrt{\pi}$):

$$\gamma_n=\sqrt{\pi}2^nn!\,.$$

The linearization coefficients $B_n(j,k)$ in the general linearization problem:

$$H_j^{(\lambda)}(x)H_k^{(\mu)}(x)=\sum_{n=0}^{\min(j,k)}B_{j+k-2n}(j,k)H_{j+k-2n}^{(\alpha)}(x)\,,\quad \alpha,\lambda,\mu>-\frac{1}{2}\,,$$

are given in [7, Eq. (3.23)] and [9, Eq. (3.5)] for the standardization of generalized Hermite polynomials $\{\mathcal{H}_n^{(\alpha)}\}_{n\geq 0}$ introduced by Rosenblum [27]:

$$\mathcal{H}_n^{(\alpha)}(x)=\frac{n!}{\gamma_\alpha(n)}H_n^{(\alpha)}(x)\,,$$

where $n\mapsto\gamma_\alpha(n)$ plays the role of a generalized factorial:

$$\begin{aligned}\gamma_\alpha(2m)&=2^{2m}m!\left(\alpha+\frac{1}{2}\right)_m\,,\\ \gamma_\alpha(2m+1)&=2^{2m+1}m!\left(\alpha+\frac{1}{2}\right)_{m+1}\,.\end{aligned}$$

Again, in the context of PC expansions we are rather interested in the standard linearization problem $\lambda=\mu=\alpha$, for which the linearization coefficients for the chosen standardization (9) read:

$$B_{j+k-2n}(j,k)=\frac{\gamma_\alpha(j)\gamma_\alpha(k)}{\gamma_\alpha(j+k-2n)}\sum_{p=0}^{\lfloor\frac{j}{2}\rfloor}\sum_{q=0}^{\lfloor\frac{k}{2}\rfloor}\frac{\gamma_\alpha(j+k-2(p+q))}{\gamma_\alpha(j-2p)\gamma_\alpha(k-2q)p!q!}\frac{(-n)_{p+q}}{n!}\,.$$

The explicit linearization formula for classical Hermite polynomials $\alpha=0$ is known as the Feldheim's formula and reads [13] (see also [9, Eq. (3.10)]):

$$H_j(x)H_k(x)=\sum_{n=0}^{\min(j,k)}\binom{j}{n}\binom{k}{n}2^n n!H_{j+k-2n}(x)\,,\tag{10}$$

where $\binom{j}{n}=\frac{j!}{(j-n)!n!}$ is the usual binomial coefficient for two integers $j\geq n$. We arrive at:

$$B_l(j,k)=\frac{\sqrt{\pi}2^n\,j!k!l!}{(n-j)!(n-k)!(n-l)!}\tag{11}$$

whenever $2n=j+k+l$ is even, and $l\leq j+k$, $k\leq j+l$, $j\leq k+l$; and $B_l(j,k)=0$ otherwise. This formula agrees with *e.g.* [5, Eq. (8)] or [32, p. 273] up to a proper normalization of the Hermite polynomials.

C. Laguerre polynomials

The generalized Laguerre polynomials $\{L_n^{(\alpha)}\}_{n\geq 0}$ are orthogonal with respect to the weight function $\mu(x)=x^\alpha\mathrm{e}^{-x}$, with $\alpha>-1$ and $I=[0,+\infty($. They are defined by *e.g.* the Rodrigues' formula:

$$L_n^{(\alpha)}(x)=\frac{1}{\mu(x)}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(\mu(x)\frac{x^n}{n!}\right)=\sum_{j=0}^{n}\binom{n+\alpha}{n-j}\frac{(-x)^j}{j!}\,,$$

and arise in gPC expansions for random variables following gamma distributions; see *e.g.* [30,37]. This family reduces to the classical Laguerre polynomials $\{L_n\}_{n\geq 0}$ for $\alpha=0$, applicable to exponentially distributed random variables. The normalization constant γ_n in Eq. (3) reads:

$$\gamma_n=\frac{(n+\alpha)!}{n!}\,.$$

The linearization coefficients $B_n(j,k)$ in the general linearization problem:

$$L_j^{(\lambda)}(x)L_k^{(\mu)}(x)=\sum_{n=|j-k|}^{j+k}B_n(j,k)L_n^{(\alpha)}(x)\,,\quad \alpha,\lambda,\mu>-1\,,$$

are given in [7, Eq. (3.24)] in terms of double hypergeometric functions. Again, in the context of PC expansions we are rather interested in the standard linearization problem $\lambda = \mu = \alpha$, for which the linearization coefficients are given by [28, 35] in terms of a terminating hypergeometric series ${}_3F_2$:

$$B_{j+k-n}(j,k) = \frac{(-2)^n}{n!}\frac{(j+k-n)!}{(j-n)!(k-n)!}\,{}_3F_2\left(\left.\begin{matrix} -\frac{n}{2},\, -\frac{n-1}{2},\, j+k-n+\alpha+1 \\ j-n+1,\, k-n+1 \end{matrix}\right| 1\right).$$

The first $\max(n-j, n-k)$ terms of the series above are ignored whenever $n > \max(j,k)$; thus:

$$B_{j+k-n}(j,k) = \frac{(-2)^n(j+k-n)!}{(j+k-n+\alpha)!n!}\sum_{l=\max(0,n-j,n-k)}^{\lfloor\frac{n}{2}\rfloor}\left(-\frac{n}{2}\right)_l\left(-\frac{n-1}{2}\right)_l\frac{(j+k-n+\alpha+l)!}{(j-n+l)!(k-n+l)!}. \qquad (12)$$

IV. Numerical implementation

The various formulas above have been implemented in `Matlab`. The routines are distributed under CeCILL-C license and are freely available at `https://github.com/ericsavin/LinCoef/`. They were compared with the results obtained with classical Gauss quadratures for the computation of the third-order moments of Eq. (4). The Golub-Welsch algorithm [18] is used for computing Gauss quadrature weights and nodes. The recurrence coefficients for monic Jacobi, generalized Hermite, and generalized Laguerre polynomials in:

$$P_{n+1}(x) = (x-c_n)P_n(x) - d_nP_{n-1}(x)$$

are given in the table 1 below, together with the leading-order coefficient p_n and the zero-th moment $\mu_0 = \langle 1\rangle_\mu$ for completeness. Computations by the analytical formulas detailed in the foregoing section are in very good agreement with Gauss quadratures, which validate our proposed codes.

	c_n	d_n	p_n	μ_0
$J_n^{(\alpha,\beta)}(x)$	$\frac{\beta^2-\alpha^2}{(2n+\alpha+\beta)(2n+\alpha+\beta+2)}$	$\frac{4n(n+\alpha)(n+\beta)(n+\alpha+\beta)}{(2n+\alpha+\beta)^2[(2n+\alpha+\beta)^2-1]}$	$\frac{1}{2^n}\binom{2n+\alpha+\beta}{n}$	$2^{\alpha+\beta+1}\frac{\alpha!\beta!}{(\alpha+\beta+1)!}$
$H_n^{(\alpha)}(x)$	0	$\frac{1}{2}[n+\alpha(1-(-1)^n)]$	2^n	$(\alpha-\frac{1}{2})!$
$L_n^{(\alpha)}(x)$	$2n+\alpha+1$	$n(n+\alpha)$	$\frac{(-1)^n}{n!}$	$\alpha!$

Table 1. Recurrence coefficients for monic Jacobi, generalized Hermite, and generalized Laguerre polynomials.

The main function is `LinCoef.m` which computes the linearization coefficients for Jacobi, Gegenbauer, generalized Hermite, and generalized Laguerre polynomials of arbitrary parameters α and β. Chebyshev polynomials (Jacobi polynomials with $\alpha = \beta = -\frac{1}{2}$) are also specifically addressed. Three routines are provided to compare the implementation with Gauss quadratures: `TestHermite.m`, `TestJacobi.m`, and `TestLaguerre.m`. These quadrature sets are constructed with the `GNodeWt.m` function, while the polynomials are evaluated at the quadrature nodes by the dedicated functions `PGHern.m`, `PJacn.m`, and `PGLagn.m`.

V. Conclusions

In this paper, we have presented the existing results for the computation of the so-called linearization coefficients for products of orthogonal polynomials of the Jacobi, generalized Hermite, and generalized Laguerre families. These coefficients correspond to the third-order moments of orthogonal polynomials, but they also serve for the computation of higher-order moments by induction. Therefore, they can be used in the intrusive and non intrusive polynomial chaos expansion methods for uncertainty quantification of engineering systems, among other possible applications. In the intrusive approach the third-order moments and beyond arise from Galerkin-type projections of the governing equations of the system models. In non intrusive approaches, these moments pertain to the computation of higher-order statistics (skewness, kurtosis and beyond) of the output quantities of interest. These results have been implemented in `Matlab` and the codes have been validated by comparison with usual Gauss quadratures. The analytical formulas we have reviewed could be used for "on-the-fly" evaluations of the moments in the course of the computations, instead of stored numerical evaluations. The present overview concerns continuous polynomials, but it can be extended to discrete polynomials alike.

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